Assignment 3 · Ground Truth

Alewife Brook Conservation Area, Cambridge MA

ett3 · Fall 2026 · field visit Fri Oct 16 2026 · pin-up Fri Oct 30 2026, 12 pm · all sheets 11 x 17

GROUP

[FIELD: names, 2 to 3 members]

[FIELD: date of print]

SHEETS

00 Cover + index

new in this packet

01 D1 · Initial patch delineation (Part A)

slot for the group's map · 3 open slots

02 D2 · Ground-truthed patch map (Part B)

from ett3-A3_D2_patch-map_v03 · 23 open slots

03 D3 · Plot, transect, plants (Part C)

from ett3-A3_D3_plot-transect-plants_v03 · 105 open slots

04 D3b · Plant drawings (Part C5)

new in this packet · 15 open slots

05 D3c · Field drawings (Part C3)

new in this packet · 8 open slots

06 S1 · Scan atlas (appendix, method)

from ett3-A3_S1_scan-atlas_v03 · 5 open slots

07 Data · proxy and field

new in this packet · 94 open slots

STATUS

Measured (scan, Polycam 10_23_2026)

transect length 8.61 m, ground profile and vegetation height every 0.25 m

strata present, zone breaks and boundary tags (hard / gradual)

ground contours 0.05 m (index 0.25 m), plot and plan views from the mesh

42 of 143 data rows have source = scan

Proxy (P), to replace with field data

*101 of 143 data rows have source = proxy: densiometer, cover, soil, compaction, litter, species*P

*patch polygons on D2, locator outline, plot centre and radius, plant species and positions*P

*file: ett3-A3_field-data_PROXY_v01.csv*P

Open slots (numbered in the review PDF)

FIELD 24

DATA 1

CONFIRM 19

P 209

total 253 on sheets 01 to 07, listed on the last page of ett3-A3_packet_review_v01.pdf

Group work still open

[FIELD: D1 map from Part A, sheet 01]

[FIELD: plant drawings P1 to P3, sheet 04]

[FIELD: hand transect sections, sheet 05]

[FIELD: field data CSV, see README_group_v01.md]

D1 · Initial Patch Delineation

Part A. Patches delineated from satellite imagery before the field trip, with field notes. Slot: the group's map goes here.

north arrow

Group's Google Earth / My Maps map goes here

[FIELD: D1 map, export at 11.6 x 9.0 in, 300 ppi, place on underlay]

scale bar

KEY (patch classes)

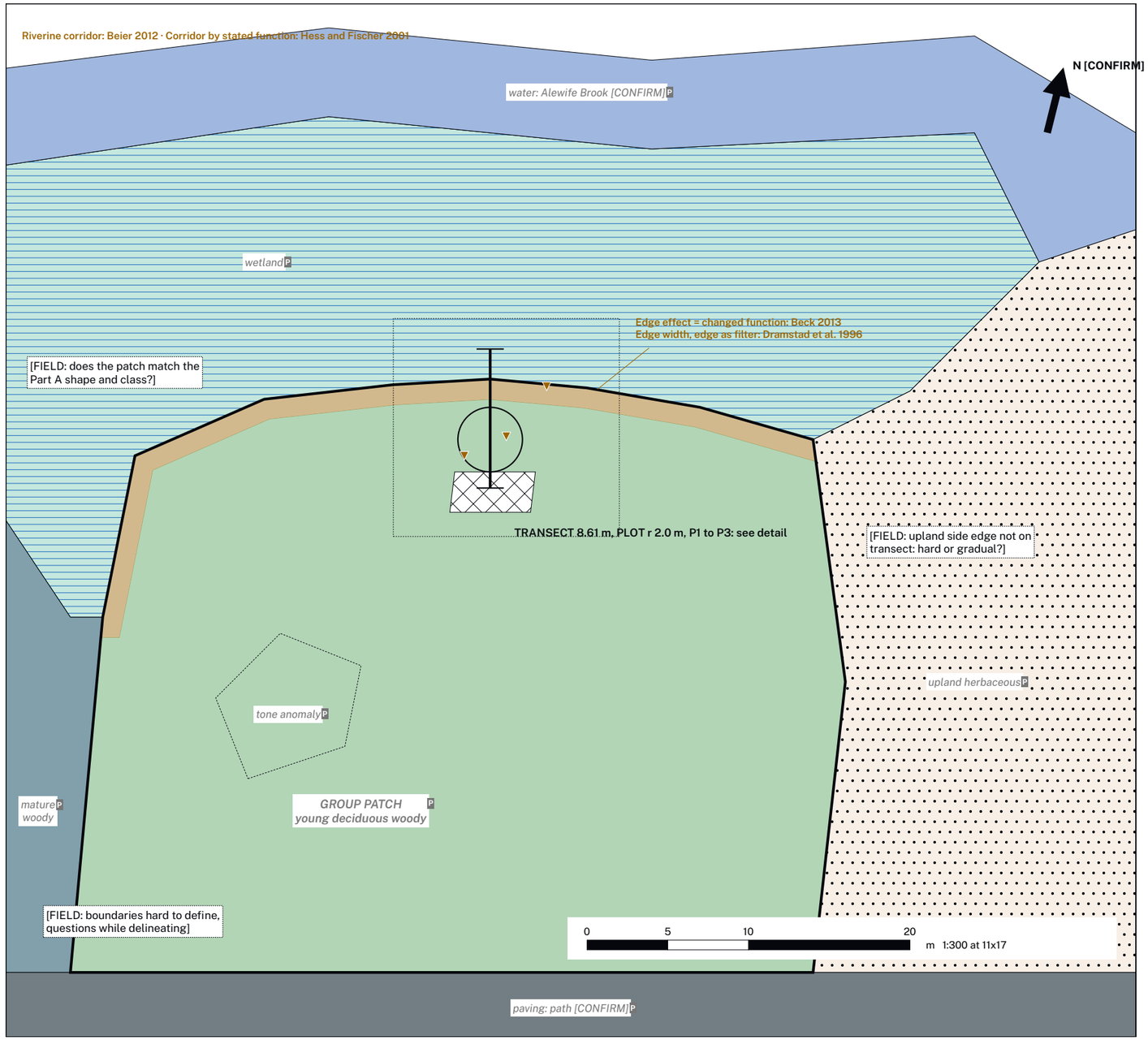
[FIELD: classes as drawn in Part A]

FIELD NOTES

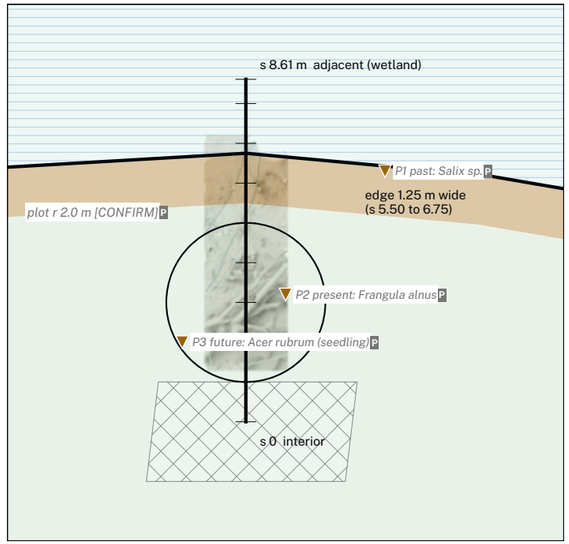
[FIELD: notes from the field trip, Oct 16]

D2 · Ground-truthed Patch Map

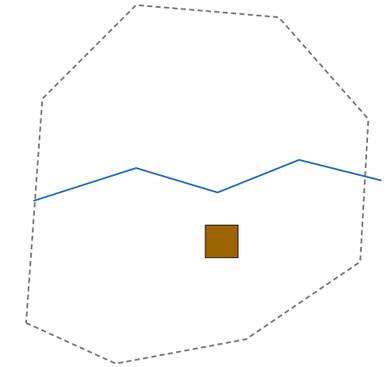
Part B. Group patch with adjacencies, edge and interior, plot, transect and plants. Patch outlines are proxy until traced from the group map.



DETAIL 1:150 · transect, plot, plants · underlay: scan mesh top view + ground contours 0.05 m



LOCATOR



[CONFIRM: trace from group KML]

KEY (spec A1 classes)

- trees / woody: young deciduous (group patch)
- trees / woody: mature deciduous
- upland herbaceous
- wetland
- bare ground / exposed soil
- paving / impervious
- water
- tone or colour anomaly in a class
- edge: functional band, 1.25 m (from transect)
- interior: patch minus edge band
- group patch boundary
- transect, ticks every 1 m
- fixed-radius plot
- P1 past · P2 present · P3 future

12.3 m measured (scan / field)

Concept: author year

[FIELD: group observation]

proxy valueP

P = proxy, replace with field data.

Classification depends on purpose: Cadenasso et al. 2013: key classes follow spec A1, refined by the transect

EDGE, DEFINED FUNCTIONALLY

Edge starts where woody height drops below 1.5 m and the bank falls 0.43 m to the wet end (scan): s 5.50 to 6.75 m.
Interior starts where woody cover holds above 1.5 m. Width 1.25 m, applied along the wetland side.

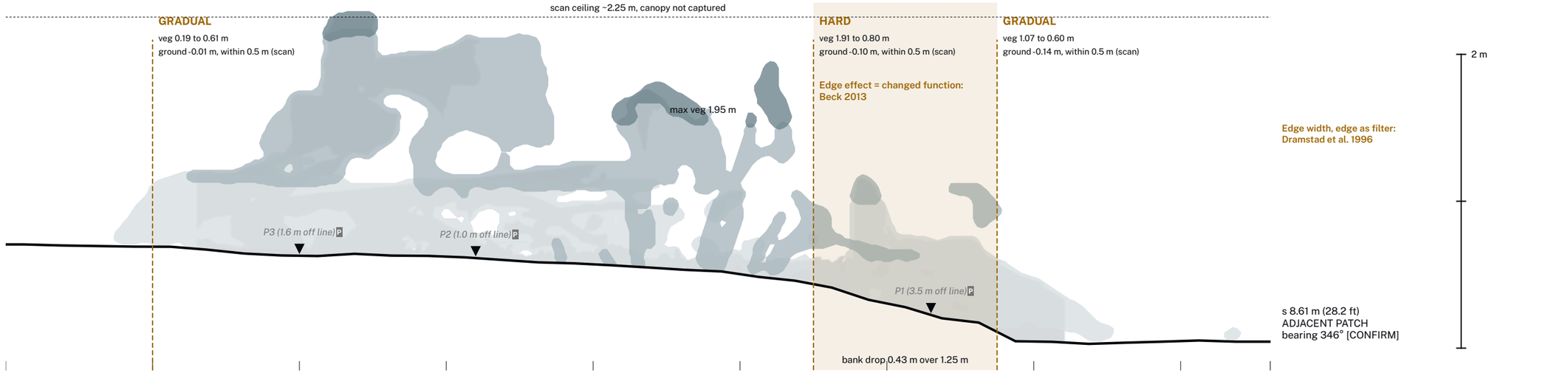
*Across it soil moisture 31 to 58 % VWC, openness 25 to 70 %, litter 4.0 to 0.5 cm*P

D3 · Plot, Transect, Plants

Part C composite. Transect in section from the Polycam scan (10_23_2026.ply), stations every 1 m, s = 0 at patch interior. Plot in plan. Plant frames for hand drawings.

SECTION H 1:30 at 11x17 · V:H 1:1 (no exaggeration) · band ±1.5 m about the axis

0 — 1 — 2 m

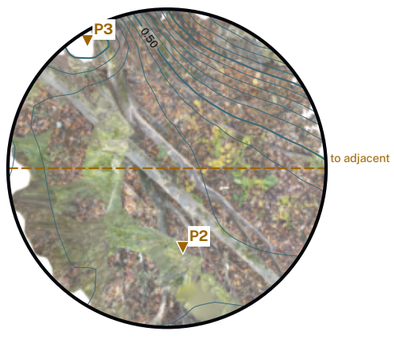


station s, m	0	1	2	3	4	5	6	7	8	8.61
zone (scan)	interior gap		interior · trees / woody, young deciduous				edge		adjacent patch	
veg height max, m (scan)	0.31	1.11	1.61	1.47	1.69	1.95	1.07	0.29	0.18	
strata present (scan)	G	G U	G U	G U	G U	G U	G U	G	G	
canopy openness, scan (scan)	no returns above 5 m at any station: scan tops at 2.25 m, canopy not captured (use densiometer row)									
canopy openness %, densiometer (proxy)	*30*P	*22*P	*18*P	*18*P	*20*P	*25*P	*40*P	*70*P	*80*P	
cover total % (proxy)	*35*P	*70*P	*85*P	*80*P	*80*P	*75*P	*60*P	*85*P	*90*P	
woody cover % (proxy)	*5*P	*55*P	*70*P	*60*P	*65*P	*55*P	*30*P	*5*P	*0*P	
species count (proxy)	*3*P	*5*P	*6*P	*6*P	*5*P	*6*P	*7*P	*5*P	*4*P	
soil moisture % VWC (proxy)	*18*P	*20*P	*22*P	*24*P	*27*P	*31*P	*42*P	*58*P	*62*P	
soil pH (proxy)	*6.4*P	*6.3*P	*6.2*P	*6.2*P	*6.1*P	*6.0*P	*6.3*P	*6.6*P	*6.7*P	
compaction kPa (proxy)	*1400*P	*900*P	*750*P	*700*P	*700*P	*650*P	*500*P	*300*P	*250*P	
litter depth cm (proxy)	*1.0*P	*3.5*P	*5.0*P	*5.5*P	*5.0*P	*4.0*P	*2.0*P	*0.5*P	*0.0*P	
dominant species (proxy)	*Poa sp.*P	*Frangula alnus*P	*Frangula alnus*P	*Frangula alnus*P	*Frangula alnus*P	*Acer rubrum*P	*Salix sp.*P	*Impatiens capensis*P	*Carex sp.*P	

strata: G ground <0.5 m · U understory 0.5 to 1.5 m · M midstory 1.5 to 5 m · O overstory >5 m (height above scan ground)

PLOT plan 1:75 · fixed radius

*r = 2.0 m, centre s = 3.0 m [CONFIRM radius]*P



TOPOGRAPHY

ground contours 0.05 m, index 0.25 m, relief 0.97 m (scan)

COMPOSITION

- *Frangula alnus 45%*P
- *Acer rubrum (seedlings) 10%*P
- *Carex sp. 15%*P
- *Toxicodendron radicans 5%*P

DENSITY / STRATA

underlay: scan texture, top view (Polycam mesh)

at s 3 m: G 77% U 18% M 0% of returns

HABITAT FEATURES

[FIELD: snags, logs, wet patches, drainages]

DISTURBANCES

[FIELD: paths, culverts, erosion, dumping, mgmt]

POINTS OF INTEREST

[FIELD: noteworthy features, impacts]

P1 · PAST

*Salix sp.*P

*at transect s 6.3 m, 3.5 m off line [CONFIRM]*P

hand drawing

Do the conditions for it exist now? Did they? Will they?

What disturbance, natural or human, helps or blocks it?

Relation to plot / transect:

P2 · PRESENT

*Frangula alnus*P

*at transect s 3.2 m, 1.0 m off line [CONFIRM]*P

hand drawing

Do the conditions for it exist now? Did they? Will they?

What disturbance, natural or human, helps or blocks it?

Relation to plot / transect:

P3 · FUTURE

*Acer rubrum (seedling)*P

*at transect s 2.0 m, 1.6 m off line [CONFIRM]*P

hand drawing

Do the conditions for it exist now? Did they? Will they?

What disturbance, natural or human, helps or blocks it?

Relation to plot / transect:

TRAJECTORY (DRAFT)

Legacy: the woody interior sits 0.66 m above the low wet end (scan); the bank, not the canopy, sets the edge. Present driver: elevation above saturated ground; moisture rises across the 1.25 m edge (P). Future: young woody growth (max 1.95 m, no stem above 5 m) thickens the interior while the wet end holds the edge in place, unless water levels shift. [FIELD: legacy plant, P1]

Shifting mosaic: Beck 2013
Legacies: Grove et al. 2015 · Time lags: Haddad et al. 2015

KEY

- ground layer <0.5 m
- understory 0.5 to 1.5 m
- midstory 1.5 to 5 m
- scan mesh, side view (whole width)
- ground (scan)
- edge zone (functional, see D2)

12.3 m measured (scan / field)

Concept: author year

[FIELD: group observation]

*proxy value*P

P = proxy, replace with field data.

D3b · Plant Drawings

Part C5. Three plants for three moments in the patch's successional trajectory. Draw by hand, scan at 300 ppi, place in the frame.

P1 · PAST

Salix sp.[P]

at transect s 6.3 m, 3.5 m off line [CONFIRM][P]

hand drawing

[FIELD: P1 drawing, scan and place]

Do the conditions for this species exist now? Did they use to? Will they?

How might disturbance, natural or human, facilitate or inhibit it?

Location in the patch; relation to plot / transect:

[FIELD: drawn by]

P2 · PRESENT

Frangula alnus[P]

at transect s 3.2 m, 1.0 m off line [CONFIRM][P]

hand drawing

[FIELD: P2 drawing, scan and place]

Do the conditions for this species exist now? Did they use to? Will they?

How might disturbance, natural or human, facilitate or inhibit it?

Location in the patch; relation to plot / transect:

[FIELD: drawn by]

P3 · FUTURE

Acer rubrum (seedling)[P]

at transect s 2.0 m, 1.6 m off line [CONFIRM][P]

hand drawing

[FIELD: P3 drawing, scan and place]

Do the conditions for this species exist now? Did they use to? Will they?

How might disturbance, natural or human, facilitate or inhibit it?

Location in the patch; relation to plot / transect:

[FIELD: drawn by]

D3c · Field Drawings

Part C3. Each member's hand section of the transect, drawn in the field: aspect, changes in plant community composition and structure.

SECTION 1 [FIELD: name] [FIELD: s from / to, facing]

hand section, scanned at 300 ppi

s 0 interior — adjacent s 8.61

SECTION 2 [FIELD: name] [FIELD: s from / to, facing]

hand section, scanned at 300 ppi

s 0 interior — adjacent s 8.61

SECTION 3 [FIELD: name] [FIELD: s from / to, facing]

hand section, scanned at 300 ppi (optional, 3rd / 4th member)

s 0 interior — adjacent s 8.61

SECTION 4 [FIELD: name] [FIELD: s from / to, facing]

hand section, scanned at 300 ppi (optional, 3rd / 4th member)

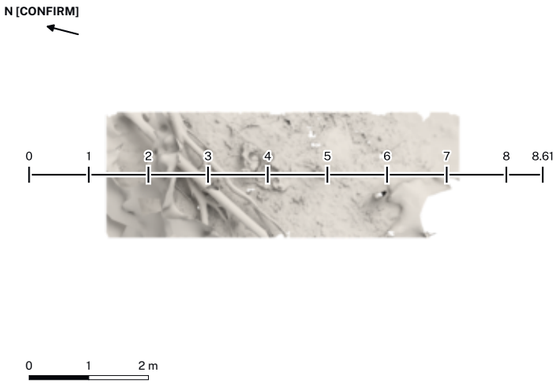
s 0 interior — adjacent s 8.61

S1 · Scan Atlas

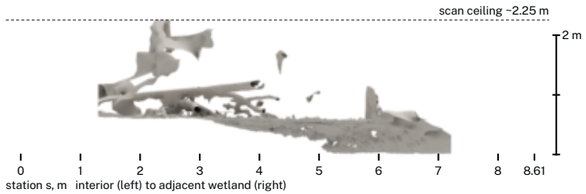
Polycam mesh of the transect (10_23_2026.zip, OBJ + baked texture), trimmed to a 5.91 x 2.10 m rectangle on the transect, in three shadings and three views. Top and side are orthographic in the transect frame, side looks across the transect. s = 0 at patch interior.

TOP ortho 1:100

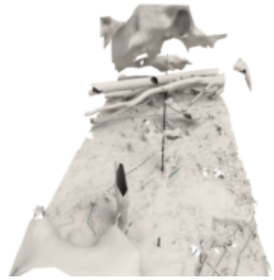
CLAY flat warm grey, studio light + AO



SIDE ortho 1:100, looking across the transect



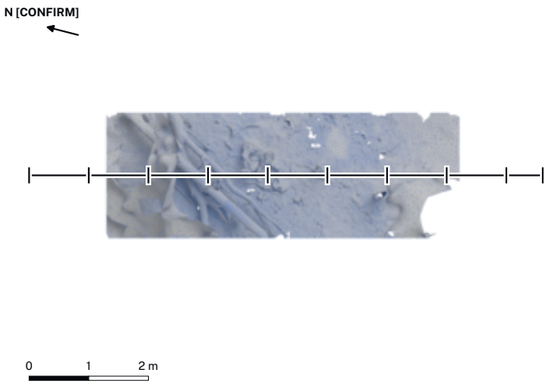
PERSPECTIVE CAM_persp_A

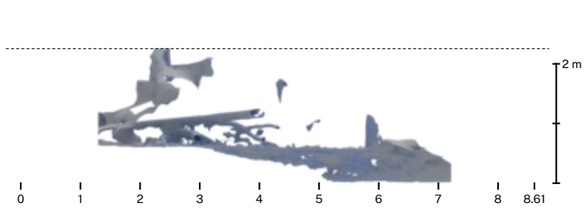


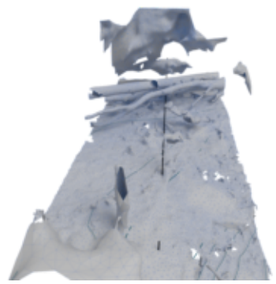
GROUND CONTOURS over the texture top view, ortho 1:75



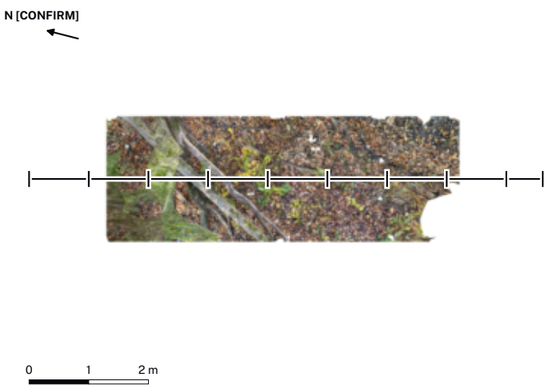
WIRE shaded grey, mesh edges in water blue







TEXTURE baked Polycam texture, unlit (no added light)







perspective, no scale (22 mm lens, 36 mm sensor). from the adjacent wetland end, looking back at the fallen log

KEY

transect on the PCA axis of the scan, ticks every 1 m (stations)

ground contours 0.05 m, index 0.25 m labelled (m, scan z)

scan ceiling: the mesh stops at 2.25 m

scan ceiling ~2.25 m, canopy not captured.

Contours are cut from the ground surface grid (5th percentile z per 0.25 m cell), not the mesh, so stems and logs do not ring.

North is not known: Polycam's frame is local. Arrow assumes scan +y = north [CONFIRM].

Mesh trimmed to a 5.91 x 2.10 m rectangle on the transect: 51 084 faces (full scan 100 000), one 4K colour texture. Rendered in Blender 4.4.1, CYCLES CPU from 01_src/blender/ett3-A3_scan_v01.blend.

Full-size views (3000 px, transparent PNG): 00_share/views/.

Measured from scan only; no proxy values on this sheet.

Data · Proxy and Field

The data file behind D2 and D3, one row per station. Grey italic + P = proxy. Replace with field data: see README_group_v01.md.

station s, m	veg ht max, m	strata	openness, %	cover, %	woody, %	species, n	moisture, % VWC	soil pH	compaction, kPa	litter, cm	dominant species	source
0	0.31	ground	*30* P	*35* P	*5* P	*3* P	*18* P	*6.4* P	*1400* P	*1.0* P	*Poa sp.* P	proxy 9 · scan 2
1	1.11	ground|understory	*22* P	*70* P	*55* P	*5* P	*20* P	*6.3* P	*900* P	*3.5* P	*Frangula alnus* P	proxy 9 · scan 2
2	1.61	ground|understory	*18* P	*85* P	*70* P	*6* P	*22* P	*6.2* P	*750* P	*5.0* P	*Frangula alnus* P	proxy 9 · scan 2
3	1.47	ground|understory	*18* P	*80* P	*60* P	*6* P	*24* P	*6.2* P	*700* P	*5.5* P	*Frangula alnus* P	proxy 9 · scan 2
4	1.69	ground|understory	*20* P	*80* P	*65* P	*5* P	*27* P	*6.1* P	*700* P	*5.0* P	*Frangula alnus* P	proxy 9 · scan 2
5	1.95	ground|understory	*25* P	*75* P	*55* P	*6* P	*31* P	*6.0* P	*650* P	*4.0* P	*Acer rubrum* P	proxy 9 · scan 2
6	1.07	ground|understory	*40* P	*60* P	*30* P	*7* P	*42* P	*6.3* P	*500* P	*2.0* P	*Salix sp.* P	proxy 9 · scan 2
7	0.29	ground	*70* P	*85* P	*5* P	*5* P	*58* P	*6.6* P	*300* P	*0.5* P	*Impatiens capensis* P	proxy 9 · scan 2
8	0.18	ground	*80* P	*90* P	*0* P	*4* P	*62* P	*6.7* P	*250* P	*0.0* P	*Carex sp.* P	proxy 9 · scan 2

ZONES (scan)

s 0.00 to 1.00 m	interior gap (bare ground / exposed soil)
s 1.00 to 5.50 m	interior (trees / woody)
s 5.50 to 6.75 m	edge (trees / woody)
s 6.75 to 8.61 m	adjacent patch (wetland)
transect length	8.61 m

PLOT

centre_s_m	*3.0 m* P
radius_m	*2.0 m* P
cover: Frangula alnus	*45 %* P
cover: Acer rubrum (seedlings)	*10 %* P
cover: Carex sp.	*15 %* P
cover: Toxicodendron radicans	*5 %* P

PLANTS

P1 past	*Salix sp.* P
position x y, m	*3.5 6.3* P
P2 present	*Frangula alnus* P
position x y, m	*1.0 3.2* P
P3 future	*Acer rubrum (seedling)* P
position x y, m	*-1.6 2.0* P

KEY

12.3 m measured (scan / field)

Concept: author year

[FIELD: group observation]

proxy value P

P = proxy, replace with field data.

[DATA: replace with field data, see README_group_v01.md]